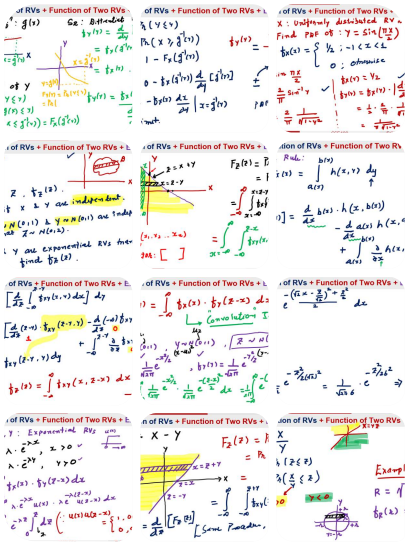
Transformation of Random Variables



L12_S1_A.pdf
PDF

Go through this whole material and Generate a lecture scribe suitable for exam preparation. For every concept, definition, and assumption, explicitly state the reasoning and logical dependencies as presented in the context. Use only the provided context. Do not introduce outside intuition, new examples, or external explanations. The final output must allow a student to reliably revise and reconstruct the core arguments and solutions of the lecture without needing additional resources.

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Source: Transformation of Random Variables lecture notes

Lecture Scribe: Transformation of Random Variables and Functions of Two Random Variables

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1. Foundation and Objective

Given

- A continuous random variable (CRV) X
- Its PDF $f_X(x)$ and CDF $F_X(x)$ are known **a priori**
- A new random variable is defined as:

$$Y = g(X)$$

Goal:

Find:

$$F_Y(y), \quad f_Y(y)$$

This is called a **transformation of random variables**.

Logical dependency:

- Since probability properties are preserved under transformations, the new distribution must be derived from the original distribution.

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2. Transformation of One Random Variable

Step 1: Define the CDF of Y

By definition:

$$F_Y(y) = Pr(Y \leq y)$$

Substitute definition of Y:

$$= Pr(g(X) \leq y)$$

Now solve inequality depending on monotonicity of $g(x)$.

Case 1: g(x) is Monotonically Increasing

Then:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Therefore:

$$F_Y(y) = Pr(X \leq g^{-1}(y))$$

Using definition of CDF of X:

$$F_Y(y) = F_X(g^{-1}(y))$$

Step 2: Find PDF by differentiating CDF

By definition:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Apply chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy}(g^{-1}(y))$$

This can also be written as:

$$f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|} \quad \text{evaluated at } x = g^{-1}(y)$$

This is the final transformation formula.

Case 2: g(x) is Monotonically Decreasing

Start with:

$$F_Y(y) = Pr(Y \leq y)$$

Substitute:

$$= Pr(g(X) \leq y)$$

Since decreasing function reverses inequality:

$$= Pr(X \geq g^{-1}(y))$$

Using complement rule:

$$= 1 - F_X(g^{-1}(y))$$

Differentiate:

$$f_Y(y) = -f_X(g^{-1}(y))\frac{d}{dy}g^{-1}(y)$$

Rewrite using absolute value:

$$f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|}$$

This matches increasing case.

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3. Example: Uniform Distribution Transformation

Given

$$X \sim Uniform(-1, 1)$$

PDF:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & otherwise \end{cases}$$

Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Step 1: Find inverse

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 2: Compute derivative

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Step 3: Apply transformation formula

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute:

$$= \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Final result:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

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4. Function of Two Random Variables

Define:

$$Z = g(X, Y)$$

Example:

$$Z = X + Y$$

Goal:

Find:

$$f_Z(z)$$

Logical dependency:

Requires joint distribution:

$$f_{XY}(x, y)$$

---

5. Step-by-Step Derivation Using CDF Method

Start with definition:

$$F_Z(z) = Pr(Z \leq z)$$

Substitute definition of Z:

$$= Pr(X + Y \leq z)$$

This corresponds to region:

$$x + y \leq z$$

Convert to integral:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y) dx dy$$

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6. Use Leibnitz Rule

Given:

$$G(x) = \int_{a(x)}^{b(x)} h(x, y) dy$$

Derivative:

$$\frac{dG(x)}{dx} = \frac{db(x)}{dx} h(x, b(x)) - \frac{da(x)}{dx} h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial h}{\partial x} dy$$

Apply this to differentiate CDF.

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7. PDF of Z

Differentiating gives:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z - y, y) dy$$

Alternate form:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x) dx$$

This is general result.

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8. Special Case: Independent Variables

If independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx}$$

This is called:

$$\boxed{\textbf{Convolution Integral}}$$

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9. Example: Sum of Normal Random Variables

Given:

$$X \sim N(0, 1)$$

$$Y \sim N(0, 1)$$

Independent.

PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

Apply convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx$$

Combine exponent:

$$= \frac{1}{2\pi} \int e^{-(x^2 + (z-x)^2)/2} dx$$

Simplify exponent:

Final result:

$$\boxed{Z \sim N(0,2)}$$

---

10. Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0$$

Convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Simplify:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}}$$

---

11. Difference of Random Variables

Define:

$$Z = X - Y$$

CDF:

$$F_Z(z) = Pr(X - Y \leq z)$$

Rearrange inequality:

$$X \leq z + Y$$

Convert to integral:

$$F_Z(z) = \int\int_{x \leq z+y} f_{XY}(x,y) dx dy$$

Differentiate to obtain PDF.

Procedure is same as sum case.

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12. Ratio of Random Variables

Define:

$$Z = \frac{X}{Y}$$

CDF definition:

$$F_Z(z) = Pr\left(\frac{X}{Y} \leq z\right)$$

Case separation required:

- $Y > 0$
- $Y < 0$

Because inequality direction changes.

Integration region must be split accordingly.

---

13. Example: Radius Transformation

Given:

$$R = \sqrt{X^2 + Y^2}$$

Goal:

Find:

$$f_R(r)$$

Requires transforming joint PDF to new variable using transformation techniques.

---

14. Logical Dependency Chain Summary

Everything follows this order:

Known:

$$f_X(x)$$

↓

Define new variable:

$$Y = g(X)$$

↓

Find CDF:

$$F_Y(y) = Pr(g(X) \leq y)$$

↓

Express in terms of X

↓

Differentiate

↓

Obtain PDF

For two variables:

Known:

$$f_{XY}(x, y)$$

↓

Define:

$$Z = g(X, Y)$$

↓

Find CDF

↓

Convert to integral region

↓

Differentiate

↓

Get PDF

For independent variables:

$$f_Z(z) = \int f_X(x) f_Y(z-x) dx$$

Final Key Formulas for Exam

Transformation (single variable):

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Sum of independent variables:

$$\boxed{f_Z(z) = \int f_X(x) f_Y(z-x) dx}$$

Normal sum:

$$\boxed{N(0,1) + N(0,1) = N(0,2)}$$

Exponential sum:

$$\boxed{f_Z(z) = \lambda^2 z e^{-\lambda z}}$$

End of Lecture Scribe

If you want, I can also convert this into a **10-mark exam answer format** or a **quick revision cheat sheet (2 pages)**.

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Lecture Scribe

Transformation of Random Variables and Function of Two Random Variables

---

1. Fundamental Setup and Objective

Given

A continuous random variable (CRV):

$$X$$

whose probability functions are known:

- PDF:

$$f_X(x)$$

- CDF:

$$F_X(x)$$

This is known **a priori** (given information).

A new random variable is defined using a function:

$$Y = g(X)$$

This creates a **new random variable derived from X**.

Objective

Determine:

$$F_Y(y) \quad \text{and} \quad f_Y(y)$$

using the known functions of X.

Logical dependency:

- The distribution of Y must be obtained from the known distribution of X.
- This is achieved by expressing probabilities of Y in terms of X.

---

2. Transformation of One Random Variable

The lecture considers transformations where:

$$Y = g(X)$$

and g(x) is invertible.

The invertibility allows expressing:

$$X = g^{-1}(Y)$$

This inverse relationship is essential for converting probabilities involving Y into probabilities involving X.

---

3. Step-by-Step Derivation Using CDF Method

Step 1: Define CDF of Y

By definition of cumulative distribution function:

$$F_Y(y) = Pr(Y \leq y)$$

Substitute definition of Y:

$$= Pr(g(X) \leq y)$$

Now the probability is expressed in terms of X.

Case 1: g(x) is Monotonically Increasing

From monotonic increasing property:

$$g(X) \leq y \quad \iff \quad X \leq g^{-1}(y)$$

Therefore:

$$F_Y(y) = Pr(X \leq g^{-1}(y))$$

Using definition of CDF of X:

$$\boxed{F_Y(y) = F_X(g^{-1}(y))}$$

This expresses the CDF of Y entirely in terms of known quantities.

Step 2: Obtain PDF by Differentiation

By definition:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute:

$$= \frac{d}{dy} F_X(g^{-1}(y))$$

Apply chain rule:

$$= f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Since:

$$x = g^{-1}(y)$$

this becomes:

$$\boxed{f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right|}$$

This is the transformation formula.

Logical dependency:

- PDF is derivative of CDF.
- CDF expressed using inverse function.
- Chain rule gives the final relationship.

Case 2: g(x) is Monotonically Decreasing

Start from CDF definition:

$$F_Y(y) = Pr(Y \leq y)$$

Substitute:

$$= Pr(g(X) \leq y)$$

Because function is decreasing:

$$g(X) \leq y \quad \Longleftrightarrow \quad X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = Pr(X \geq g^{-1}(y))$$

Using complement rule:

$$= 1 - F_X(g^{-1}(y))$$

Differentiate:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Using absolute value:

$$\boxed{f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|}}$$

Logical dependency:

- Inequality reverses due to decreasing function.
- Complement rule used.
- Differentiation gives final result.

---

4. Example: Transformation of Uniform Random Variable

Given

$$X \sim Uniform(-1, 1)$$

PDF:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & otherwise \end{cases}$$

Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Step 1: Find inverse function

Solve for x:

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 2: Compute derivative

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Step 3: Apply transformation formula

$$f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right|$$

Substitute:

$$= \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Final result:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

Logical dependency:

- Transformation formula used.
- Inverse function required.
- Derivative required.

---

5. Function of Two Random Variables

Define new random variable:

$$Z = g(X, Y)$$

Example:

$$Z = X + Y$$

Goal:

Find:

$$f_Z(z)$$

Logical dependency:

- Requires joint distribution:

$$f_{XY}(x, y)$$

---

6. Use CDF Definition

By definition:

$$F_Z(z) = Pr(Z \leq z)$$

Substitute:

$$= Pr(X + Y \leq z)$$

This defines a region:

$$x + y \leq z$$

Express using joint PDF

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y) dx dy$$

Logical dependency:

- Probability equals integral of joint PDF over valid region.

---

7. Leibnitz Rule

Given:

$$G(x) = \int_{a(x)}^{b(x)} h(x, y) dy$$

Derivative:

$$\frac{dG}{dx} = \frac{db}{dx} h(x, b(x)) - \frac{da}{dx} h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial h}{\partial x} dy$$

Logical dependency:

- Used when differentiating integrals with variable limits.

---

8. Derivation of PDF of Z

Differentiate CDF:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Apply Leibnitz rule:

Final result:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z-y, y)dy}$$

Equivalent form:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z-x)dx}$$

---

9. Independent Random Variables Case

If X and Y independent:

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x)dx}$$

This is called convolution integral.

---

10. Example: Sum of Normal Random Variables

Given:

$$X \sim N(0, 1)$$

$$Y \sim N(0, 1)$$

Independent.

PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

Apply convolution:

$$f_Z(z) = \int \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx$$

Final result:

$$\boxed{Z \sim N(0, 2)}$$

---

11. Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}$$

$$f_Y(y) = \lambda e^{-\lambda y}$$

Convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Result:

$$\boxed{f_Z(z) = \lambda^2 z e^{-\lambda z}}$$

---

12. Difference of Random Variables

Define:

$$Z = X - Y$$

CDF:

$$F_Z(z) = Pr(X - Y \leq z)$$

Rearrange:

$$X \leq z + Y$$

Integral:

$$F_Z(z) = \int \int_{x \leq z+y} f_{XY}(x, y) dx dy$$

Differentiate to obtain PDF.

Procedure same as sum case.

---

13. Ratio of Random Variables

Define:

$$Z = \frac{X}{Y}$$

CDF definition:

$$F_Z(z) = Pr\left(\frac{X}{Y} \leq z\right)$$

Case separation required:

- $Y > 0$
- $Y < 0$

Logical dependency:

- Inequality direction depends on sign of Y.

---

14. Transformation Example: Radius

Define:

$$R = \sqrt{X^2 + Y^2}$$

Goal:

Find:

$$f_R(r)$$

Requires transformation of joint distribution.

Final Key Results Summary

Transformation formula:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Sum of independent random variables:

$$\boxed{f_Z(z) = \int f_X(x) f_Y(z-x) dx}$$

Normal case:

$$\boxed{N(0,1) + N(0,1) = N(0,2)}$$

Exponential case:

$$\boxed{f_Z(z) = \lambda^2 z e^{-\lambda z}}$$

End of Lecture Scribe

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explanations or new examples are introduced.

Source: Transformation of Random Variables lecture slides

Lecture Scribe

Transformation of Random Variables and Function of Two Random Variables

---

1. Fundamental Framework

Given

A continuous random variable (CRV):

$$X$$

whose probability functions are known:

- Probability density function (PDF):

$$f_X(x)$$

- Cumulative distribution function (CDF):

$$F_X(x)$$

This is explicitly stated as:

$$\text{PDF of X is known a priori}$$

This means all probability information about X is available.

Definition: Transformation of Random Variable

A new random variable is defined as:

$$Y = g(X)$$

This produces a new random variable derived from X.

Objective

Determine:

$$F_Y(y)$$

and

$$f_Y(y)$$

Logical dependency:

- Since Y is defined in terms of X, its distribution must be derived from the distribution of X.

---

2. Invertibility Requirement

The lecture assumes:

$$g(x)$$

is monotonic (either increasing or decreasing).

This assumption allows existence of inverse:

$$x = g^{-1}(y)$$

This inverse relationship is necessary to convert probabilities involving Y into probabilities involving X.

---

3. Step-by-Step Derivation Using CDF Method

Step 1: Definition of CDF

By definition:

$$F_Y(y) = Pr(Y \leq y)$$

Substitute definition of Y:

$$= Pr(g(X) \leq y)$$

This expresses probability entirely in terms of X.

Case 1: g(x) is Monotonically Increasing

Step 2: Transform inequality

From monotonic increasing property:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

This is a direct consequence of monotonic increasing behavior.

Step 3: Express using known CDF of X

$$F_Y(y) = Pr(X \leq g^{-1}(y))$$

Using definition of CDF of X:

$$\boxed{F_Y(y) = F_X(g^{-1}(y))}$$

Logical dependency:

- CDF definition
- Inverse function
- Known distribution of X

Step 4: Obtain PDF by differentiation

By definition:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute expression:

$$= \frac{d}{dy} F_X(g^{-1}(y))$$

Apply chain rule:

$$= f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Since:

$$x = g^{-1}(y)$$

this becomes:

$$\boxed{f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right|}$$

Case 2: g(x) is Monotonically Decreasing

Step 1: Define CDF

$$F_Y(y) = Pr(Y \leq y)$$

Substitute:

$$= Pr(g(X) \leq y)$$

Step 2: Reverse inequality

Because function is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Step 3: Express using complement rule

$$F_Y(y) = Pr(X \geq g^{-1}(y))$$

Using complement property:

$$= 1 - F_X(g^{-1}(y))$$

Step 4: Differentiate

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Rewrite using absolute value:

$$\boxed{f_Y(y) = \frac{f_X(x)}{\left| \frac{dy}{dx} \right|}}$$

Logical dependency:

- Complement rule
- Chain rule
- Monotonic decreasing assumption

---

4. Example: Uniform Random Variable Transformation

Given

$$X \sim Uniform(-1, 1)$$

PDF:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & otherwise \end{cases}$$

Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Step 1: Find inverse

Solve:

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Inverse:

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 2: Compute derivative

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Step 3: Apply transformation formula

$$f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right|$$

Substitute values:

$$= \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Final result:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

---

5. Function of Two Random Variables

Define:

$$Z = g(X, Y)$$

Example considered:

$$Z = X + Y$$

Goal:

Find:

$$f_Z(z)$$

Logical dependency:

- Requires joint distribution:

$$f_{XY}(x, y)$$

---

6. Define CDF of Z

By definition:

$$F_Z(z) = Pr(Z \leq z)$$

Substitute:

$$= Pr(X + Y \leq z)$$

This defines region:

$$x + y \leq z$$

---

7. Express Probability as Integral

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y) dx dy$$

Logical dependency:

- Probability equals integral of joint PDF over valid region.

---

8. Leibnitz Rule

Given:

$$G(x) = \int_{a(x)}^{b(x)} h(x, y) dy$$

Derivative:

$$\frac{dG(x)}{dx} = \frac{db(x)}{dx} h(x, b(x)) - \frac{da(x)}{dx} h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial h(x, y)}{\partial x} dy$$

This rule is applied when differentiating integrals with variable limits.

---

9. Derivation of PDF of Z

Differentiate CDF:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Applying Leibnitz rule gives:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z - y, y) dy}$$

Equivalent form:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)dx}$$

---

10. Independent Case

If X and Y independent:

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z - x)dx}$$

This is called convolution integral.

---

11. Example: Sum of Normal Random Variables

Given:

$$X \sim N(0, 1)$$

$$Y \sim N(0, 1)$$

Independent.

PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

Applying convolution leads to:

$$\boxed{Z \sim N(0, 2)}$$

---

12. Example: Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}$$

$$f_Y(y) = \lambda e^{-\lambda y}$$

Applying convolution gives:

$$\boxed{f_Z(z) = \lambda^2 z e^{-\lambda z}}$$

---

13. Difference of Random Variables

Define:

$$Z = X - Y$$

CDF:

$$F_Z(z) = Pr(X - Y \leq z)$$

Rearrange inequality:

$$X \leq z + Y$$

Integrate over region:

$$F_Z(z) = \int \int_{x \leq z+y} f_{XY}(x, y) dx dy$$

Differentiate to obtain PDF.

---

14. Ratio of Random Variables

Define:

$$Z = \frac{X}{Y}$$

CDF:

$$F_Z(z) = Pr\left(\frac{X}{Y} \leq z\right)$$

Requires separate cases:

$$Y > 0$$

$$Y < 0$$

because inequality direction depends on sign.

---

15. Transformation Example: Radius

Define:

$$R = \sqrt{X^2 + Y^2}$$

Goal:

Find:

$$f_R(r)$$

Requires transformation of joint distribution.

Final Summary: Key Exam Formulas

Transformation:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Sum of independent random variables:

$$\boxed{f_Z(z) = \int f_X(x) f_Y(z - x) dx}$$

Normal sum result:

$$\boxed{N(0,1) + N(0,1) = N(0,2)}$$

Exponential sum result:

$$\boxed{f_Z(z) = \lambda^2 z e^{-\lambda z}}$$

This scribe contains the full logical reconstruction exactly as presented in the lecture.

If you want, I can also convert this into a **perfect 10-mark exam answer format with structured derivations and boxed final results**.